

Smart City:

Energy Consumption Analytics

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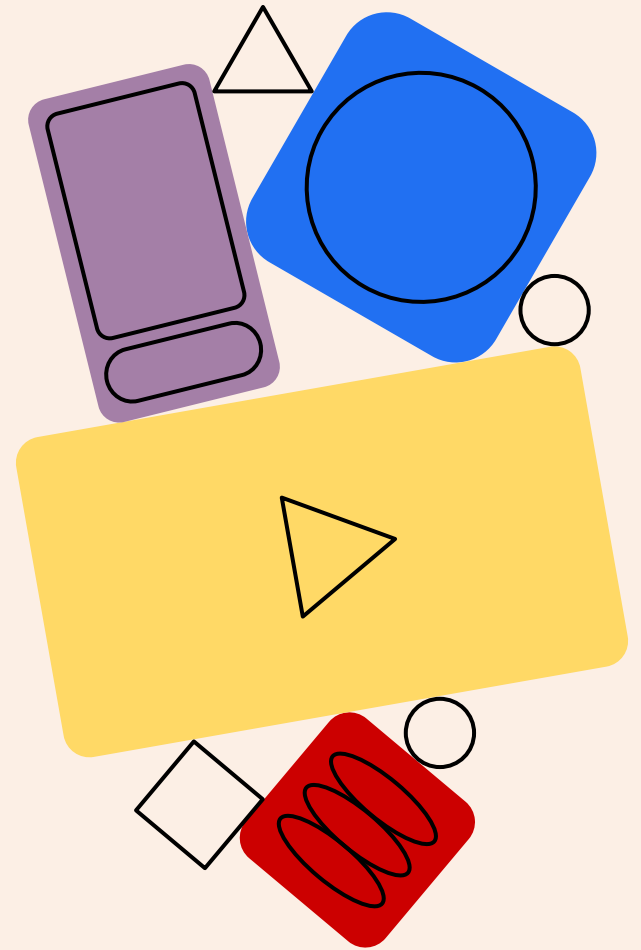


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BUSINESS IMPACT

Business Problem

Problem Statement

Modern cities generate enormous amounts of electricity consumption data every hour.

However,

- Energy usage varies across residential, commercial and industrial properties.
- Authorities lack a centralized analytical platform.
- Identifying high energy consumption zones is difficult.
- Peak electricity demand is unpredictable.
- Renewable energy adoption needs continuous monitoring.
- Decision-making is often based on fragmented operational data.



Need

A centralized analytics solution capable of monitoring, analysing and visualizing city-wide energy consumption.

Project Overview

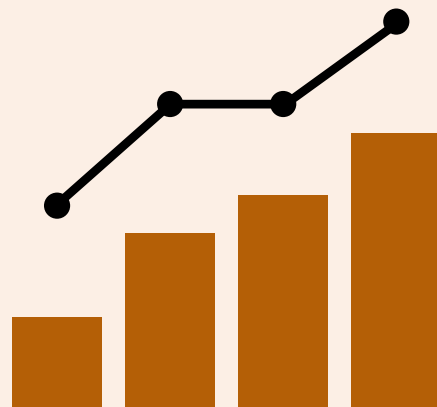
Goal

Develop a Business Intelligence solution that transforms operational energy data into meaningful insights.

Solution	• Oracle Database as Data Warehouse • Oracle Analytics Cloud (OAC) • Star Schema Design • SQL-based analytical views • Interactive dashboards
Outcome	• Smart city planning • Sustainable energy management • Renewable energy analysis • Evidence-based decision making

Business Objectives

- Monitor electricity consumption across city zones.
- Analyse residential, commercial and industrial energy usage.
- Identify high consumption regions.
- Study hourly and seasonal consumption trends.
- Measure renewable energy adoption.
- Calculate CO₂ offset generated by renewable sources.
- Support strategic planning through interactive dashboards.
- Enable data-driven decisions for efficient energy distribution.



Workflow & Architecture



01

DATA SOURCES

- Property Information
- Zone Information
- Energy Source Details
- Date & Time Information



02

ORACLE DATABASE

Oracle FreeSQL Database, where SQL queries are executed to prepare datasets for Oracle Analytics Cloud (OAC).



03

ORACLE ANALYTICS CLOUD

Provides interactive dashboards and visualizations to analyze energy consumption trends and support data-driven decision-making.



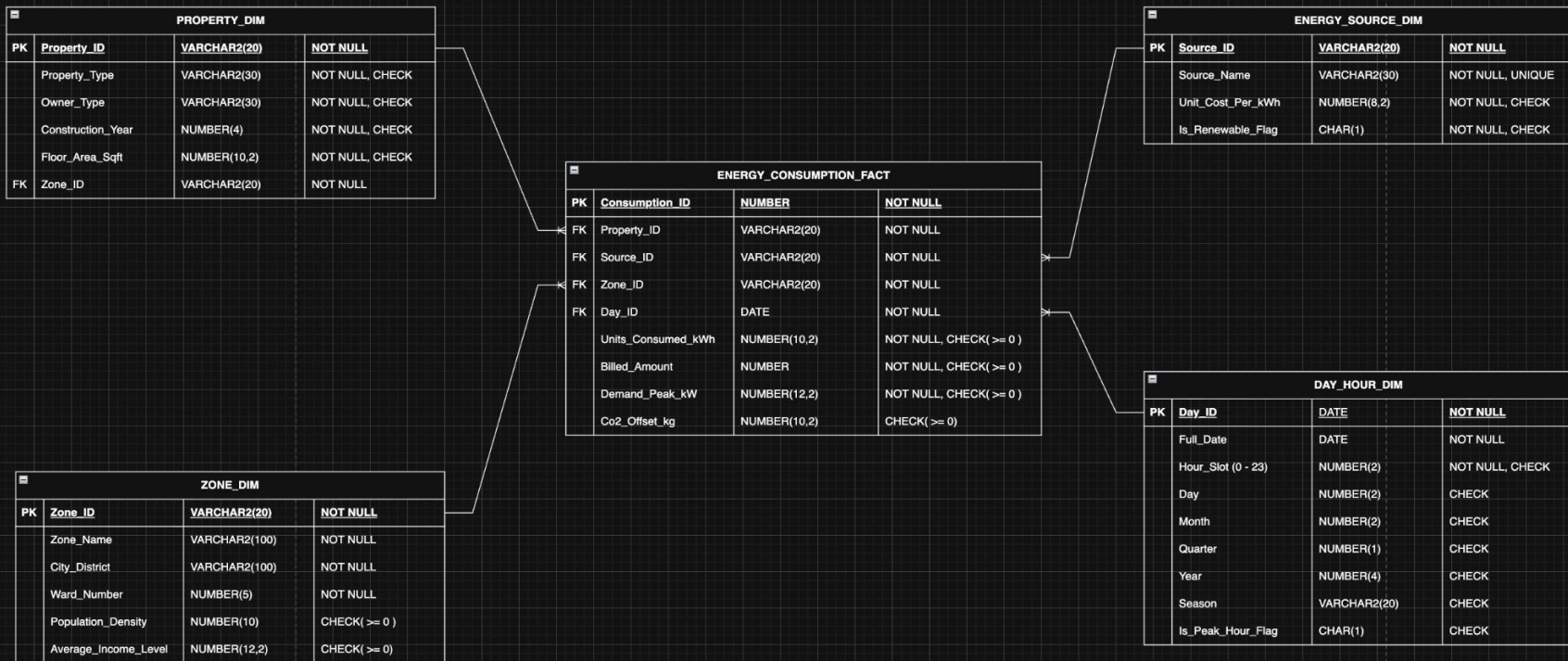
04

USERS

- City Authorities
- Energy Analysts
- Policy Makers



SCHEMA DESIGN



Granularity

(Hour Level Data Storage)



The data warehouse stores energy consumption at an hourly level, enabling detailed analysis of usage patterns, peak demand periods, seasonal trends, and more accurate business insights.

Benefits

- Identifies peak electricity demand hours
- Enables detailed hourly trend analysis
- Supports seasonal consumption analysis
- Improves forecasting and decision-making
- Provides more accurate analytical insights

Trade-off

- Requires more storage than daily-level data, but delivers significantly richer and more precise analytics.

Business Deliverables

#1

Which Zone and Property type combination has the highest average energy consumption per month ?

#2

What percentage of total energy consumption is sourced from renewable resources, broken down by zone and quarter?

#3

Which hour slots consistently have the highest peak demand (kW) across the city ?

#4

What is the total CO₂ Offset achieved through renewable energy usage across all zones in last year ?

Business Deliverable 01

Monthly Energy Consumption Analysis

Dashboard shows

- Average monthly consumption
- Zone-wise comparison
- Property-type comparison
- Consumption trends



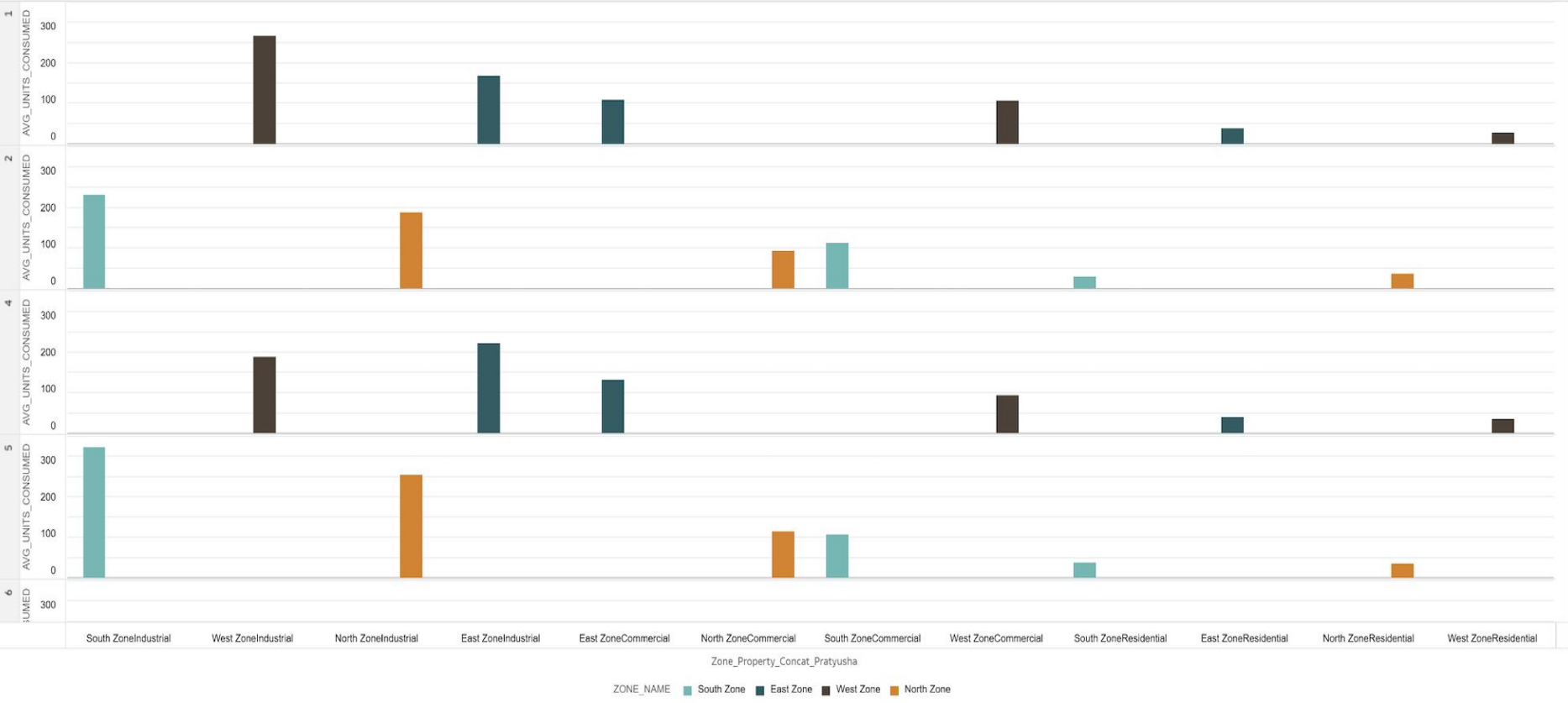
Business Insight

Helps identify

- High consumption zones
- Industrial energy demand
- Seasonal usage patterns



AVG_UNITS_CONSUMED by Zone_Property_Concat_Pratyusha, ZONE_NAME, MONTH



Business Deliverable 02

Renewable Energy Analysis

Measures

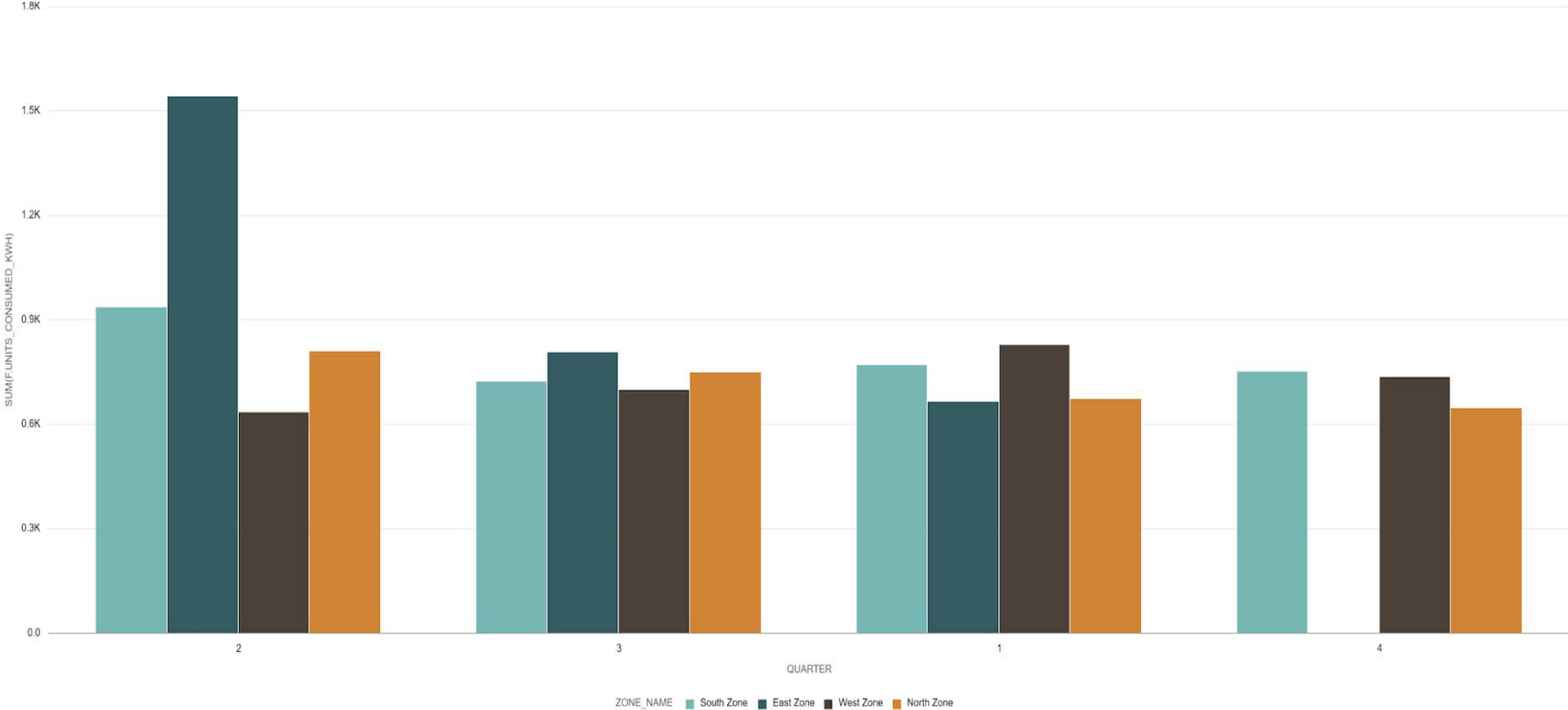
- Renewable energy units
- Total energy units
- Renewable percentage
- Quarter-wise analysis
- Zone-wise comparison

Business Value

- Measure clean energy adoption
- Track sustainability goals
- Promote renewable investments



SUM(F.UNITS_CONSUMED_KWH) by QUARTER, ZONE_NAME



Business Deliverable 03

Peak Demand Analysis

Displays

- Average peak demand
- Maximum demand
- Hour-wise comparison

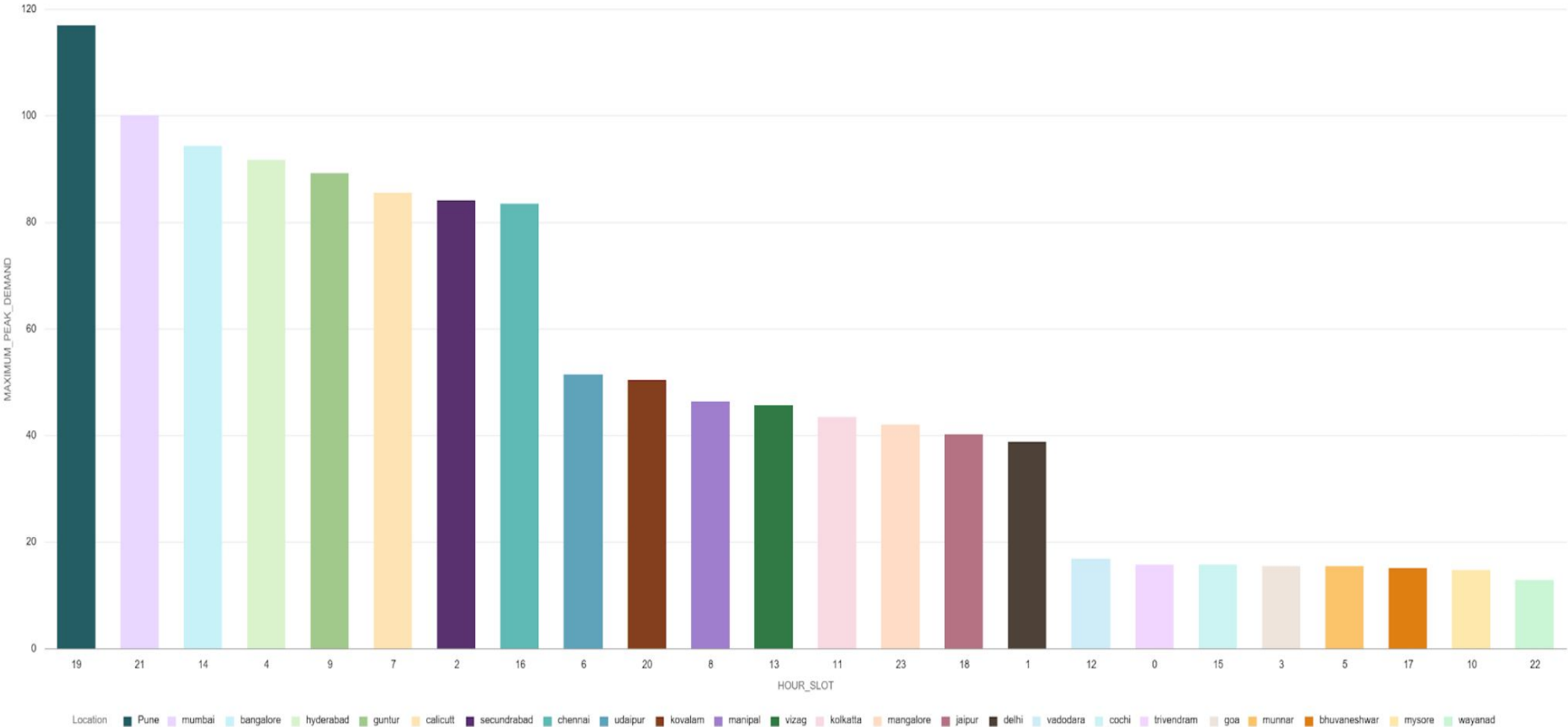


Business Value

- Identify busiest hours
- Improve load balancing
- Support electricity distribution planning
- Reduce power shortages



MAXIMUM_PEAK_DEMAND by HOUR_SLOT, Location



Business Deliverable 04

CO₂ Offset Analysis

Shows

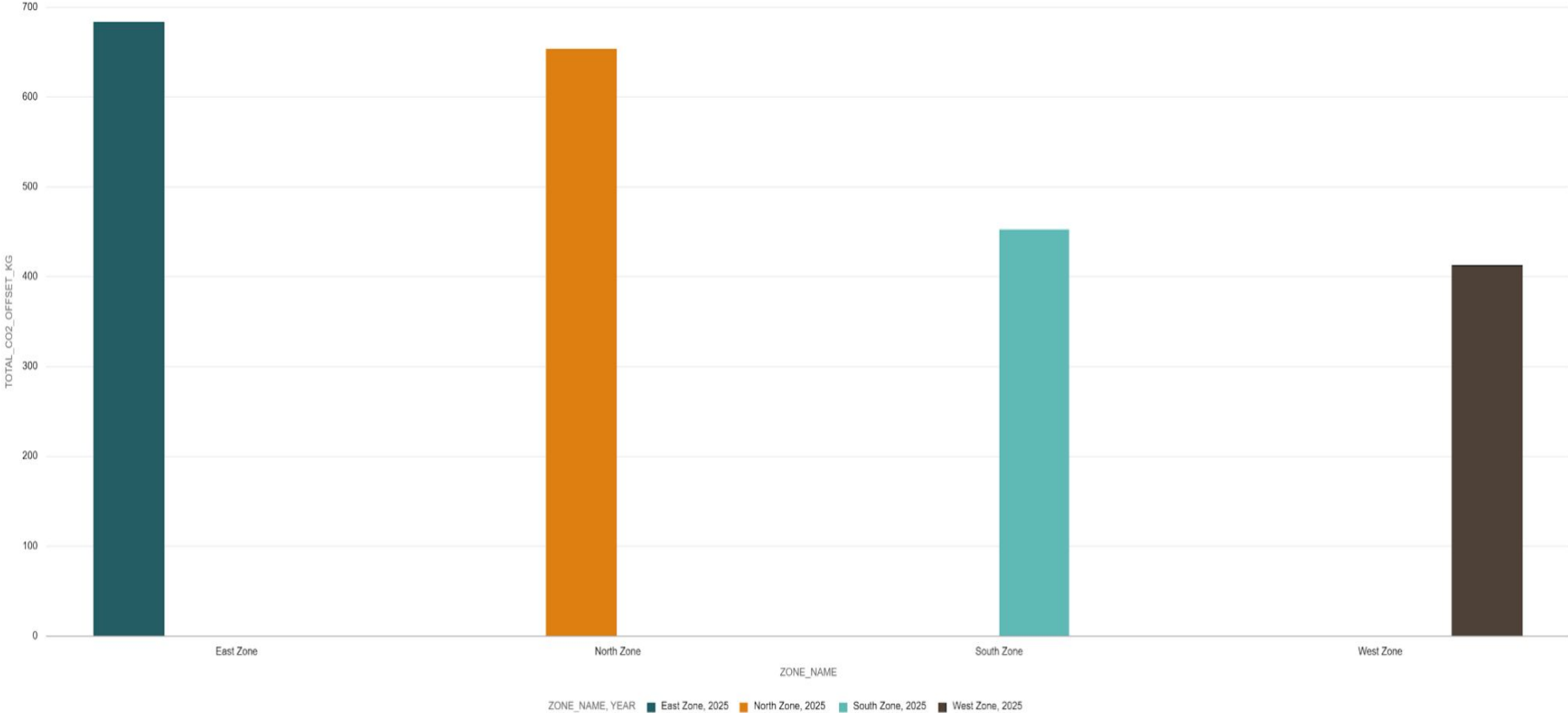
- Total CO₂ offset
- Zone-wise comparison
- Year-wise renewable contribution

Benefits

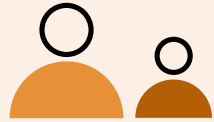
- Measure environmental impact
- Evaluate renewable initiatives
- Support green city planning



TOTAL_CO2_OFFSET_KG by ZONE_NAME, YEAR



Business Impact



Operational Benefits

- Centralized analytics
- Faster reporting
- Reduced manual analysis

Strategic Benefits

- Better resource allocation
- Smart energy planning
- Renewable energy monitoring
- Peak demand prediction

Environmental Benefits

- Lower carbon emissions
- Sustainable city development
- Improved energy efficiency